

BI Dashboard & Data Visualization

Global Superstore Sales & Profitability Dashboard

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1. INTRODUCTION

1.1 Background

Modern organizations collect large volumes of transactional data every day, but raw transactional records on their own do not support decision-making. The value of this data is only realized when it is cleaned, modelled, and presented in a form that allows managers to quickly identify patterns, risks, and opportunities. Business Intelligence (BI) dashboards address this problem by converting raw data into interactive, visual analytical solutions.

1.2 Project Objective

This project focuses on developing an interactive Business Intelligence dashboard using the Global Superstore dataset. The objective is to transform transactional sales data into meaningful business information through data cleaning, data modelling, DAX calculations, interactive visualizations, and dashboard-based analysis.

The dashboard enables users to analyse:

- Sales, profit, customers, and orders at a glance (KPI-level monitoring)
- Monthly and yearly sales and profit trends
- Product performance and Top 10 products by revenue
- Category and sub-category performance (with drill-down)
- Regional profitability across 13 global regions
- The relationship between discount levels and profitability
- Individual product behaviour through drill-through analysis

1.3 Scope

The solution was built entirely in Microsoft Power BI Desktop, covering the full BI workflow: raw CSV ingestion → Power Query transformation → star-schema data model → DAX measures → a five-page interactive dashboard (Executive Overview, Product Analysis, two Profitability Analysis pages, and a Product Details drill-through page).

2. DATASET & METHODOLOGY

2.1 Dataset Description

The Global Superstore dataset contains transactional information related to customer orders, products, sales, discounts, profit, shipping, regions, and dates. The dataset was imported into Power BI and transformed using Power Query before creating the analytical data model.

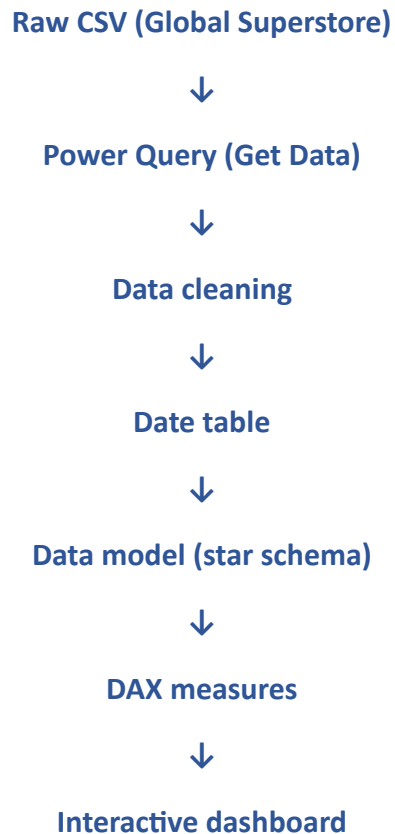
Key fields used in the analysis:

Field Group	Example Fields	Used For
Order information	Order ID, Order Date, Ship Date, Ship Mode	Trend analysis, time intelligence
Customer information	Customer ID, Customer Name, Segment (Consumer / Corporate / Home Office)	Customer KPIs, segmentation
Geography	Region (Central, South, North, Oceania, EMEA, Africa, etc.), Country, City, Market	Regional profitability
Product information	Product ID, Product Name, Category, Sub-Category	Product & category analysis
Measures	Sales, Quantity, Discount, Profit, Shipping Cost	KPI cards, profitability, discount analysis

The data covers approximately 25,000 orders placed by thousands of customers between 2011 and 2014, generating roughly \$12.64M in sales and \$1.47M in profit.

2.2 Data Preparation Pipeline

The following workflow was followed from raw data to the finished dashboard:



2.3 Main Transformations Performed

In Power Query (data cleaning):

- Imported the raw CSV file and promoted the first row to headers
- Verified and corrected data types (dates, decimal numbers, whole numbers, text)
- Removed duplicate records and blank/erroneous rows
- Trimmed text fields and standardised column names
- Handled missing values in non-critical fields
- Derived Year and Month columns from the Order Date field to support trend analysis

In the data model:

- Created a dedicated Date table covering 2011–2014 with Year, Quarter, Month, and Month Number columns, and marked it as the official date table

- Built a star schema: a central fact table (order transactions) related to dimension tables for Dates, Products, Customers, and Geography using one-to-many, single-direction relationships
- Used the Date table as the axis for all time-based visuals, enabling consistent year-over-year and monthly analysis

In DAX (calculated measures):

Measure	Example Definition	Purpose
Total Sales	Total Sales = SUM(Orders[Sales])	Headline revenue KPI
Total Profit	Total Profit = SUM(Orders[Profit])	Headline profit KPI
Total Orders	Total Orders = DISTINCTCOUNT(Orders[Order ID])	Order-volume KPI
Total Customers	Total Customers = DISTINCTCOUNT(Orders[Customer ID])	Customer-base KPI
Profit Margin %	DIVIDE([Total Profit], [Total Sales])	Profitability ratio
Average Order Value	DIVIDE([Total Sales], [Total Orders])	Spending behaviour
Average Discount	AVERAGE(Orders[Discount])	Discount-profitability analysis

3. DASHBOARD DESIGN

The final report consists of five interactive pages grouped into four functional areas, connected by navigation buttons displayed on every page, with a subtitle banner: *"Executive Overview / Sales • Profit • Customers • Orders."* Global slicers (Year, Category, Segment, Ship Mode, Region) are synced across pages so that a filter selection carries through the entire analysis, and a Reset Filters button restores the default view.

3.1 Page 1 — Executive Overview

The landing page provides an at-a-glance summary of business health:

- Six KPI cards — Total Sales (\$12.64M), Total Profit (\$1.47M), Total Orders (25K), Total Customers (5K), Profit Margin (11.6%), and Average Order Value (\$505.01)
- Sales & Profit Trend Over Time — a combination chart showing monthly sales as columns and profit as a line across 2011–2014, with data labels on each month
- Sales & Profit by Category — a clustered bar chart (Technology, Furniture, Office Supplies) supporting drill-down into sub-categories
- Slicer panel: Year dropdown, Category, Segment, Ship Mode, and Region

3.2 Page 2 — Product Analysis

- Regional Sales & Profitability — a horizontal bar chart comparing Total Sales and Total Profit across 13 regions (Central, South, North, Oceania, Southeast Asia, North Asia, EMEA, Africa, Central Asia, West, East, Caribbean, Canada)
- Top 10 Products by Sales — a ranked bar chart of the highest-revenue products within the selected filters (led by Apple Smart Phone, Full Size at \$87K)

3.3 Page 3 — Profitability Analysis (two tabs)

- Tab 1 — Discount vs Profitability: a bubble/scatter chart plotting Average Discount (X) against Total Profit (Y), with bubble size representing volume and colour representing category. A dashed reference line marks the overall average discount of 8.19%
- Tab 2 — Product Performance Analysis: a detailed table listing Product Name, Total Sales, Total Quantity, Average Discount, and Total Profit, sorted by profit, with conditional formatting (blue for profitable, red for loss-making items)

3.4 Page 4 — Product Details (Drill-through)

A focused drill-through page activated by right-clicking any product in the Profitability Analysis table:

- Product name card (e.g., *Novimex Executive Leather Armchair, Red*)

- Four KPI cards: Total Sales (\$38.813K), Total Profit (\$1.450372K), Total Quantity (118), Profit Margin (3.7%)
- Total Sales and Total Profit by Year line chart (2011–2014)
- A BACK button returning the user to the source page

4. KEY BUSINESS INSIGHTS

4.1 Insights from the Displayed (2014-Filtered) Analysis

1. Category performance Technology is the strongest-performing category in the displayed analysis, generating approximately \$117K in sales and \$19K in profit.
2. Monthly performance Monthly sales vary considerably, with November reaching approximately \$38K while February records approximately \$6.7K in the displayed 2014 analysis — a nearly six-fold difference between the best and worst months.
3. Top products Canon Wireless Fax, Laser is the leading product in the displayed Top 10, generating approximately \$5.7K in sales.
4. Highly profitable product Canon Wireless Fax, Laser generates approximately \$2.78K in profit with a 49.0% profit margin in the detailed product analysis — demonstrating how low-discount, high-margin items drive disproportionate profit.
5. Discount and losses Several products with 50% discounts show negative profitability. Nokia Smart Phone, Full Size records approximately –\$918 profit and a –36.9% profit margin in the filtered view.
6. High sales do not always mean high profit The selected Novimex Executive Leather Armchair, Red generates approximately \$38.8K in sales but only \$1.45K in profit, producing a 3.7% profit margin. Its yearly trend shows sales recovering strongly by 2014, but profit remains thin.

4.2 Additional Insights from the Full-Dataset View

7. Overall business performance Across all years, the business achieves a \$12.64M revenue base with only an 11.6% overall profit margin and an average order value of \$505.01 — indicating significant revenue leakage relative to sales volume.
8. Category margins differ sharply In the full view: Technology (\$4.7M sales / ~\$0.7M profit, ~15% margin), Office Supplies (\$3.8M / ~\$0.5M, ~13%), Furniture (\$4.1M / ~\$0.3M, ~7%). Furniture generates the second-highest sales but roughly half the margin of the other categories.

9. Regional profitability is highly uneven Central is the largest region (\$2.82M sales / \$0.31M profit). However, Southeast Asia (\$0.88M / \$0.02M, ~2% margin) and EMEA (\$0.81M / \$0.04M, ~5% margin) produce substantial revenue with minimal profit, while Canada is the smallest market (\$0.07M).

10. The most profitable product is not the top seller In the full Product Performance view, Canon imageCLASS 2200 Advanced Copier delivers the highest profit (\$25,199) on \$61,600 sales with a 0% discount, while Apple Smart Phone, Full Size — the #1 revenue product (\$86,936) — yields only \$5,921 profit (~6.8% margin).

11. Loss-making items exist even among top sellers The top-20 table contains loss-makers, including Motorola Smart Phone, Cordless (–\$4,447), Hoover Stove, White (–\$2,181), and Samsung Smart Phone, Cordless (–\$198 despite \$48.7K in sales).

12. A clear discount–profitability threshold In the Discount vs Profitability scatter, profitable products cluster below roughly 20% average discount; items beyond 40–50% discounts increasingly fall below zero, with the largest loss bubble (approximately –\$9K) occurring near a 67% average discount. Sales made at ~100% discount contribute essentially no profit.

5. BUSINESS RECOMMENDATIONS

Recommendation 1 — Review high-discount products Products with high discounts and negative profits (e.g., Nokia Smart Phone, Full Size at 50% discount; Motorola Smart Phone, Cordless at –\$4,447) should be reviewed to reduce unnecessary margin losses. Discount caps or approval thresholds should be introduced for discounts above ~20–30%.

Recommendation 2 — Focus on profitable products Products with both strong sales and high profit margins — such as Canon imageCLASS 2200 Advanced Copier (0% discount, \$25.2K profit) and Canon Wireless Fax, Laser (49% margin) — should receive appropriate attention in inventory and sales planning, marketing priority, and stock availability guarantees.

Recommendation 3 — Monitor category profitability Management should evaluate categories using sales, profit, and profit margin together rather than sales alone. Furniture, in particular, delivers high revenue at roughly half the margin of Technology and Office Supplies and warrants pricing or cost-structure review.

Recommendation 4 — Improve demand planning Large monthly variations (November ≈ \$38K vs February ≈ \$6.7K in the displayed analysis) suggest that inventory and promotional planning should account for seasonal peaks and troughs, aligning stock levels and campaigns with the November demand surge.

Recommendation 5 — Investigate weak-margin regions Southeast Asia (~2% margin) and EMEA (~5% margin) convert sales into profit poorly. A review of shipping costs, local discounting practices, and product mix in these regions is recommended.

Recommendation 6 — Separate revenue performance from profitability performance The drill-through analysis (e.g., Novimex Executive Leather Armchair, Red: \$38.8K sales, 3.7% margin) shows that sales-rank dashboards alone can mask unprofitable behaviour. Product scorecards should always pair revenue with margin.

6. CONCLUSION

The Global Superstore BI dashboard converts transactional sales data into an interactive analytical solution. Through KPIs, trend analysis, category and regional comparisons, product analysis, discount-profit analysis, and drill-through functionality, the dashboard provides a comprehensive view of business performance.

The analysis demonstrates the importance of evaluating both revenue and profitability when making business decisions: high-selling products can be low-margin or loss-making, deep discounting destroys profitability, and category and regional performance vary widely. By combining a clean star-schema data model, reusable DAX measures, and multi-page interactive design, the project shows how a structured BI workflow enables evidence-based decision-making.

Future enhancements could include sales forecasting with Power BI analytics features, customer-level profitability (RFM) analysis, a what-if parameter simulating discount policy changes, and row-level security for regional managers.